Arista Access Node HSR-ACC-27 QoS Congestion Event Troubleshooting SOP

Node: HSR-ACC-27 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_3575

Description

This Standard Operating Procedure (SOP) is designed to troubleshoot and resolve Quality of Service (QoS) congestion events on Arista Access Node HSR-ACC-27, located at HSR Layout Access. QoS congestion events occur when microbursts or interface tail drops cause QoS queue starvation, leading to packet loss and poor network performance. On Arista devices, QoS is implemented using the Modular QoS CLI (MQC), which allows for fine-grained control over traffic classification, policing, and shaping. In this context, the failure domain is specific to the Access role, where the node is configured to aggregate and forward traffic to the distribution layer. The alarm code ALARM_3575 indicates a minor severity congestion event, which may be caused by a variety of factors, including misconfigured QoS policies, insufficient buffer sizes, or hardware limitations. This SOP will guide the troubleshooter through the diagnostic and resolution process, utilizing a combination of CLI commands and network analysis tools to identify and fix the root cause of the congestion event.

Pre-requisite Checks

- Verify that the HSR-ACC-27 node is reachable and accessible via the network.

- Check the redundant link status to ensure that backup links are functioning properly.

- Dispatch remote hands to HSR Layout Access to perform physical checks and gather additional information.

Potential Root Causes

Cause	Probability
Misconfigured QoS policies	High
Insufficient buffer sizes	Medium
Hardware limitations	Low

Diagnostic Commands & Expected Outputs

```
Command: show mpls l2mc interface, Expected Output: Interface: HSR-ACC-27 (GigabitEthernet1/0/1) - L2MC status: Up, Description: This command displays the L2MC interface status, which can help identify if the congestion event is related to L2MC traffic.

Command: show mpls l2mc statistics, Expected Output: L2MC statistics: Packets sent: 1000, Packets received: 1000, Description: This command displays L2MC statistics, which can help identify if there are any issues with L2MC traffic.
```

```
Command: show mpls l2mc congestion, Expected Output: Congestion event detected: Interface HSR-ACC-27 (GigabitEthernet1/0/1), Description: This command displays congestion event information, which can help identify the root cause of the congestion event.
```

Resolution Steps

Step 1: Step: Verify QoS policies, Command: show mpls l2mc qos policy, Description: Verify that the QoS policies are correctly configured and that there are no misconfigured policies causing the congestion event.

Step 2: Step: Adjust buffer sizes, Command: mpls l2mc interface buffer-size GigabitEthernet1/0/1 10000, Description: Adjust the buffer sizes to ensure that there is sufficient buffer space to handle the traffic.

Step 3: Step: Upgrade hardware, Command: system upgrade force, Description: Upgrade the hardware to the latest version to resolve any hardware limitations causing the congestion event.

Post-Check Verification

```
Command: show mpls l2mc interface, Expected Output: Interface: HSR-ACC-27 (GigabitEthernet1/0/1) - L2MC status: Up, Description: Verify that the L2MC interface status is up and that the congestion event has been resolved.

Command: show mpls l2mc statistics, Expected Output: L2MC statistics: Packets sent: 1000, Packets received: 1000, Description: Verify that the L2MC statistics are within normal limits and that there are no issues with L2MC traffic.
```

Roll-back Procedures

- Step: Restore previous QoS policies, Command: mpls l2mc qos policy restore, Description: Restore the previous QoS policies to ensure that there are no changes made during the troubleshooting process.

- Step: Reset buffer sizes, Command: mpls l2mc interface buffer-size GigabitEthernet1/0/1 1000, Description: Reset the buffer sizes to the default values to ensure that there is no impact on the network.